Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2343116	SQL Lab	--	2	--	--	--	1	1

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II					
2343116	SQL Lab	--	--	--	--	--	25	25	50

Lab Objective: The course aims to

Sr. No	Lab Objectives
1	To identify and define problem statements for real life applications.
2	To construct conceptual data model for real life applications.
3	To Build Relational Model from ER/EER and demonstrate usage of relational algebra.
4	To Apply SQL to store and retrieve data efficiently.
5	To implement database connectivity using JDBC.
6	To understand the concepts of transaction processing- concurrency control & recovery procedures.

Lab Outcomes: Upon completion of the course, students will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Define problem statement and Construct the conceptual model for real life application.	L1, L3, L4, L6
2	Create and populate a RDBMS using SQL.	L3, L4
3	Formulate and write SQL queries for efficient information retrieval	L3, L4
4	Apply view, triggers and procedures to demonstrate specific event handling.	L1, L3, L4
5	Demonstrate database connectivity using JDBC.	L3
6	Demonstrate the concept of concurrent transactions.	L3, L4

Prerequisite: C and Python Programming.

Hardware & Software Requirements:

Hardware Requirement:	Software requirement:
PC i3 processor and above	Any SQL Compiler, Python/Java Programming Language

DRAFT COPY

DETAILED SYLLABUS:

Sr. No.	Experiment List	Hours	LO Mapping
1.	Identify real world problem and develop the problem statement. Design an Entity-Relationship (ER) / Extended Entity-Relationship (EER) Model.	**02**	LO1
2.	Mapping ER/EER to Relational schema model.	**02**	LO1
3.	Create a database using DDL and apply integrity constraints.	**02**	LO2, LO3
4.	Perform data manipulations operations on populated database.	**02**	LO3
5.	Perform Authorization using Grant and Revoke.	**02**	LO2, LO3
6.	Implement Basic and complex SQL queries.	**02**	LO3, LO4
7.	Implementation of Views and Triggers.	**02**	LO4
8.	Demonstrate database connectivity using JDBC.	**02**	LO5
9.	Execute TCL commands.	**02**	LO4
10.	Implement functions and procedures in SQL	**02**	LO3, LO4
11.	Implementation of Cursor.	**03**	LO3, LO4
12.	Implementation and demonstration of Transaction and Concurrency control techniques using locks.	**03**	LO6

DRAFT COPY

Note: Guidelines for the conduction of practical.

- **Faculty will assign one real-world case study or application to a group of 3 students, and each group is to perform the above list of experiments and then apply to their assigned case study or application.**
- **Learner must prepare a Journal of the above experiment list along with a report of their assigned case study/application.**

Text Books:

1. Korth, Slberchatz, Sudarshan, Database System Concepts, 6th Edition, McGraw Hill
2. Elmasri and Navathe, Fundamentals of Database Systems, 6th Edition, Pearson education
3. Raghu Ramkrishnan and Johannes Gehrke, Database Management Systems, TMH

References:

1. Peter Rob and Carlos Coronel, ― Database Systems Design, Implementation and Management‖, Thomson Learning, 9th Edition.
2. SQL & PL / SQL for Oracle 11g Black Book, Dreamtech Press
3. G. K. Gupta : "Database Management Systems", McGraw – Hill

Online References:

Sr. No.	Website Name
1.	https://www.nptel.ac.in
2.	https://www.oreilly.com
3.	https://www.coursera.org/

Assessment :
Term Work: Term work shall consist of 10-12 practical's based on above list. Also Term work Journal must include at report of assigned case study/application as an assignment.

Term Work Marks: 25 Marks (Total marks) = 15 Marks (Experiment) + 5 Marks (Report of Case Study/Application) + 5 Marks (Attendance)

Practical & Oral Exam: An Oral & Practical exam will be held based on the above list of 10-12 Practicals.

DRAFT COPY